Population Growth and CO_2 Emissions: A Data Analysis in R

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Abstract

This project studies how population size is related to carbon dioxide (CO_2) emissions using two Kaggle datasets: the World Population Dataset (234 countries and territories) and the CO_2 Emissions Around the World dataset (215 country records). The data were checked and cleaned, the population of each country in 2030 was projected with the exponential growth model $N_t = Pe^{rt}$, and the population and CO_2 trends of the ten most populous countries were visualised. Correlation analysis and regression were used to link population in 2022 to CO_2 emissions in 2019 for the 173 countries that could be matched by name, and emissions were compared across continents. CO_2 emissions are measured per person (metric tons per capita), not as total national emissions. The results show almost no relationship between population size and CO_2 per person (Pearson $r = 0.01$, Spearman $\rho = -0.08$, regression $R^2 = 0.01$, $p = 0.19$). Asia had the highest continental total in 2019 (266.18), North America was third (79.57) and South America was last (29.42). Population size alone therefore does not explain CO_2 emissions per person.

1 Introduction

Population growth increases the need for energy, transport, goods and other resources, which can contribute to CO_2 emissions. However, emissions also depend on how countries produce energy, their economic activities, technology and lifestyles. This project asks whether countries with larger populations also tend to have higher CO_2 emissions.

2 Methods

Datasets. The population dataset contains, for each country or territory, the continent, the population in selected years (1970–2022), area, density, growth rate and world population percentage. The CO_2 dataset gives annual CO_2 emissions in metric tons per capita from 1990 to 2019. The two were merged by country name. The CO_2 file contains the year 2019 twice; the two values were averaged so that each country has one value per year, and a few country names were harmonised (for example "Russian Federation" to "Russia").

Cleaning and EDA. Using the tidyverse packages in RStudio, I displayed the variable names, the first 5 and last 10 rows, the data types and the dimensions. Duplicate rows were dropped and missing values counted (numeric gaps would be filled with the median and text gaps with "Unknown", but none were found). Outliers were found with box plots and the IQR rule. Since they are mostly real countries (China, India, Monaco), they were kept, and a capped copy (limits at 1.5 times the IQR) was used for the correlations.

Projection. The growth rate is a yearly multiplier, so $r = \ln(\text{Growth Rate})$, P is the 2022 population and $t = 2030 - 2022 = 8$ years.

Analysis. Graphs were made with ggplot2. I computed Pearson and Spearman correlations (with a heatmap) for Area, Density, Growth rate and World population percentage. For the merged data (2022 population and 2019 CO_2) I used a log-scale scatter plot, correlations and the regression $\ln(CO_2) = a + b\ln(\text{Population})$. CO_2 was summed by continent.

3 Results

Data quality. The population dataset has 234 rows and 17 columns, with no duplicate rows and no missing values, so no imputation was needed. The IQR rule flagged many outliers: 25 to 32 in each population column, 28 in area, 22 in density, 25 in world population percentage and only 3 in growth rate.

Projection to 2030. The projection was computed for all 234 countries. For example, Afghanistan is projected to grow from 41,128,771 (2022) to 50,385,848 in 2030 and Angola from 35,588,987 to 45,610,989, while Albania, with a growth rate below 1, declines from 2,842,321 to 2,746,004.

Top ten countries (Figures 1 and 2). The ten most populous countries in 2022 were China, India, United States, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia and Mexico. China and India are far ahead of the others, with about 1.4 billion people each. Between 1990 and 2022 all ten grew, India fastest in absolute terms (from about 870 million, almost reaching China), while Russia stayed almost flat. For CO_2 per person from 1990 to 2019, the United States stayed highest but fell from about 19 to about 15 tons, Russia dropped sharply in the 1990s

and then recovered to about 12, China rose steadily from about 2 to about 7.6, and the other countries remained below 4 tons.

Correlations (Figures 2 and 3). Among the population variables, the strongest correlation is between Area and World population percentage ($r = 0.70$ Pearson, 0.84 Spearman); Area and Density are negatively correlated ($r = -0.41$), and Growth rate is only weakly related to the other variables. For the 173 matched countries, the Pearson correlation between population and CO_2 is 0.015, the Spearman correlation is -0.081 and the log–log correlation is -0.101. In the regression, the slope is -0.067 ($p = 0.186$) with $R^2 = 0.010$: population explains about 1% of the variation in CO_2 per person, and the slope is not significant.

Continents, 2019 (Table 1, Figure 3). Asia is first with a total of 266.18 metric tons per capita, North America is third with 79.57 and South America is last with 29.42.

Rank	Continent	Total CO_2 2019 (metric tons per capita)
1	Asia	266.18
2	Europe	241
3	North America	79.57
4	Africa	60.5
5	Oceania	53.2
6	South America	29.42

Table 1: Total 2019 CO_2 by continent (sum of per-capita values of the matched countries; values for Europe, Africa and Oceania are rounded as printed in the R output).

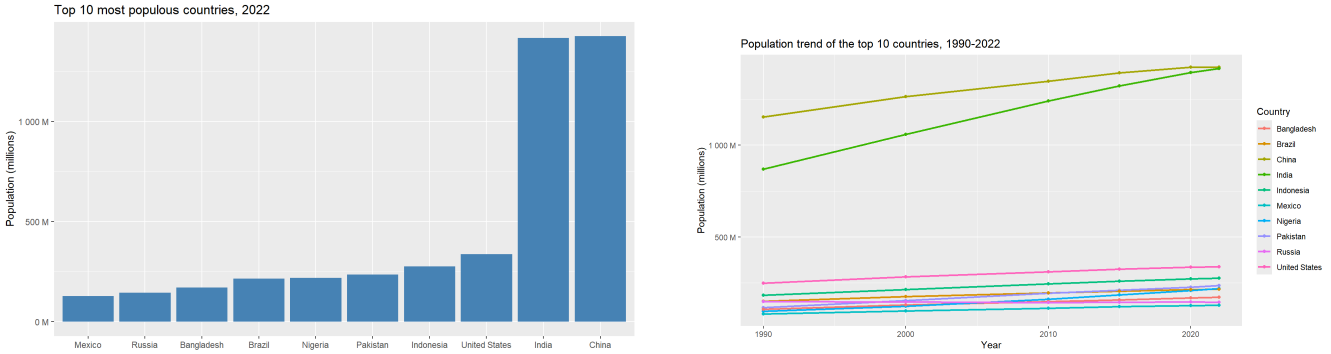


Figure 1: Top 10 most populous countries in 2022 (left) and their population trend, 1990–2022 (right).

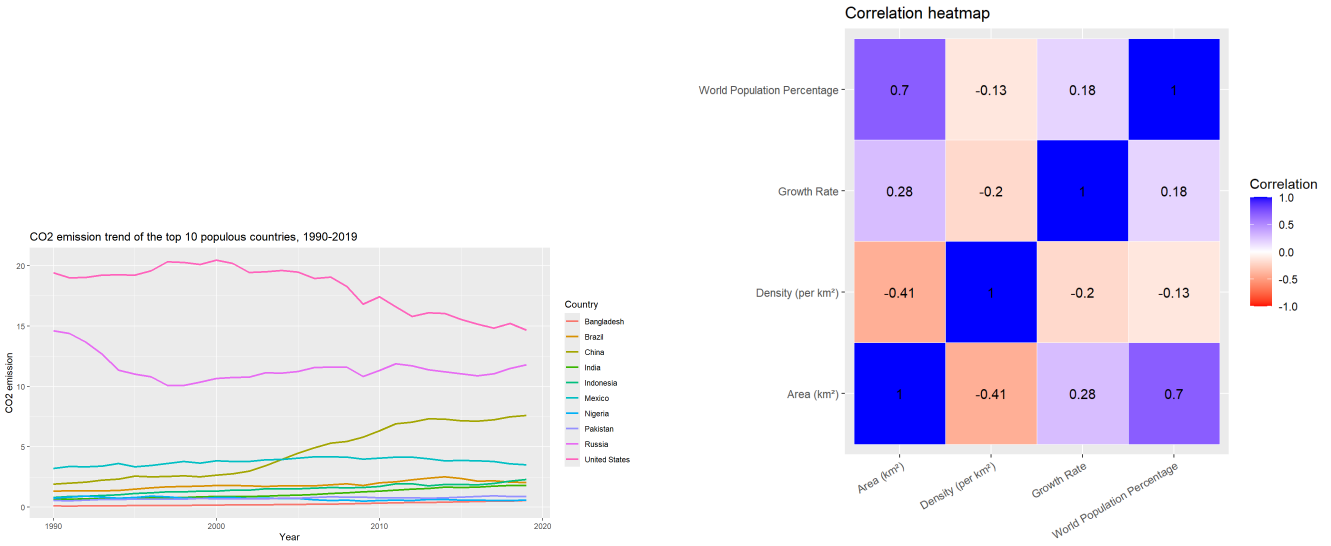


Figure 2: CO_2 trend of the 10 most populous countries, 1990–2019 (left), and correlation heatmap of the population variables (right).

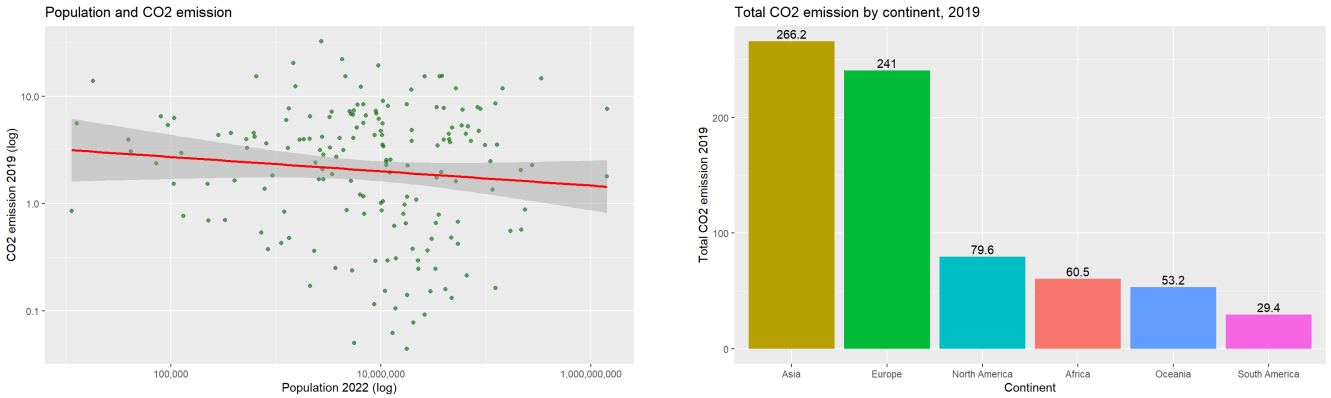


Figure 3: Population (2022) versus CO_2 (2019), log scale (left), and total 2019 CO_2 by continent (right).

4 Discussion

The main finding is that population size and CO_2 per person are essentially unrelated in these data: the correlations are close to zero and the regression is not significant. This is logical, because CO_2 here is measured per person, so a large population does not automatically mean more emissions per head; the United States, with a moderate population, emits far more per person than China or India. Income, industry and energy sources therefore matter more. Outliers have a strong influence on the raw Pearson value, which is why the Spearman correlation and the log scale were also used. The continent totals add per-capita values of countries, so they do not give the real total emission of a continent, and the ranking also depends on how many countries each continent has. Finally, only 173 of the 234 population records could be used after the merge, because some names differ (for example the Bahamas, Hong Kong and DR Congo) or CO_2 data are missing.

5 Lessons learned and recommendations

Cleaning the data and matching country names took as much effort as the analysis; outliers must be understood before being removed; and duplicated columns (like the repeated 2019) must be handled before reshaping. I recommend standardising country names before merging, using total emissions (population $\times$ per-capita value) or population-weighted averages for continent comparisons, and adding income (GDP) and energy mix to explain emissions better.

6 Conclusion

The data were cleaned, the 2030 population of each country was projected, and the ten most populous countries were analysed with their population and CO_2 trends. Correlation, regression and a continent comparison show that population size is not related to CO_2 emissions per person in these data, so other factors must explain them. Asia has the highest continental total in 2019 and South America the lowest.